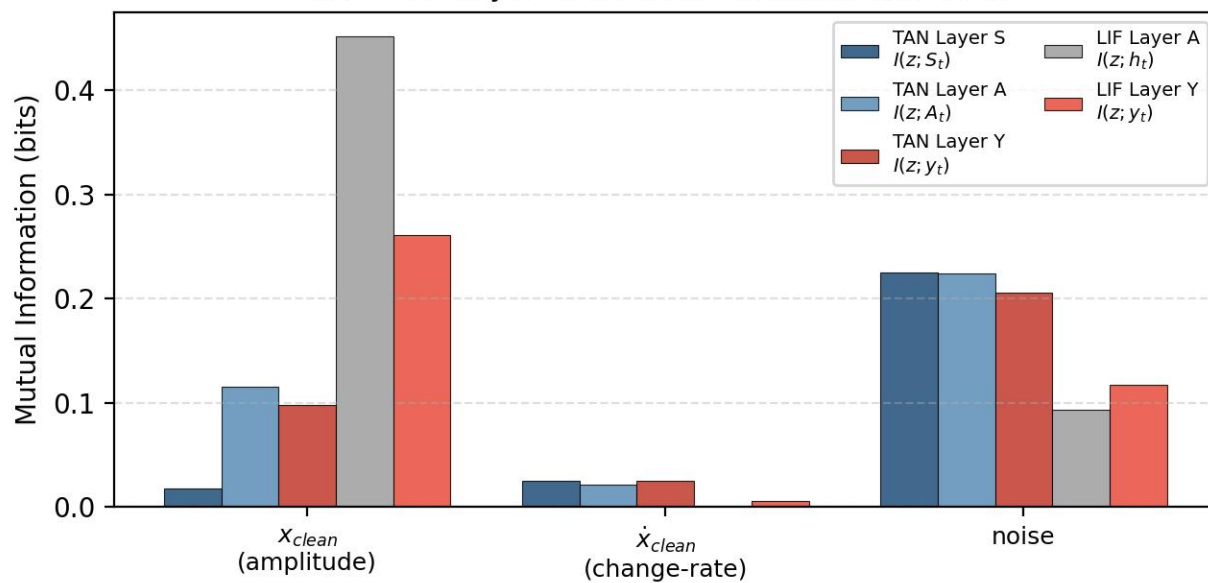


Layer 4 — Three-Layer Information Cascade Analysis

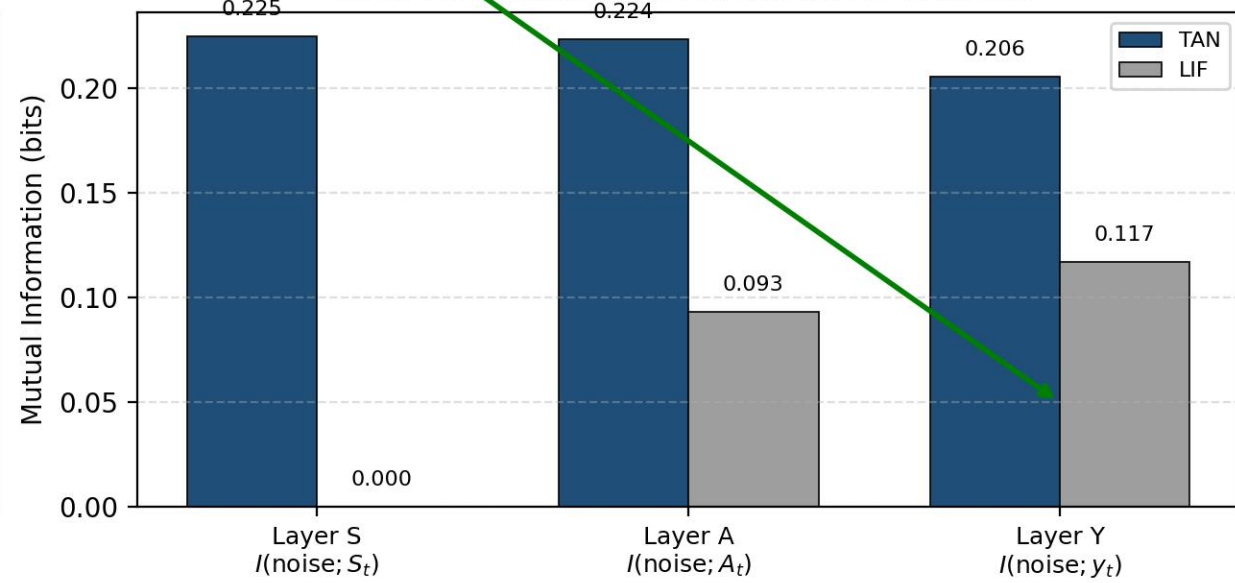
TAN as a hierarchical filter: Surprise detects → Attention relays → Threshold filters

filtering

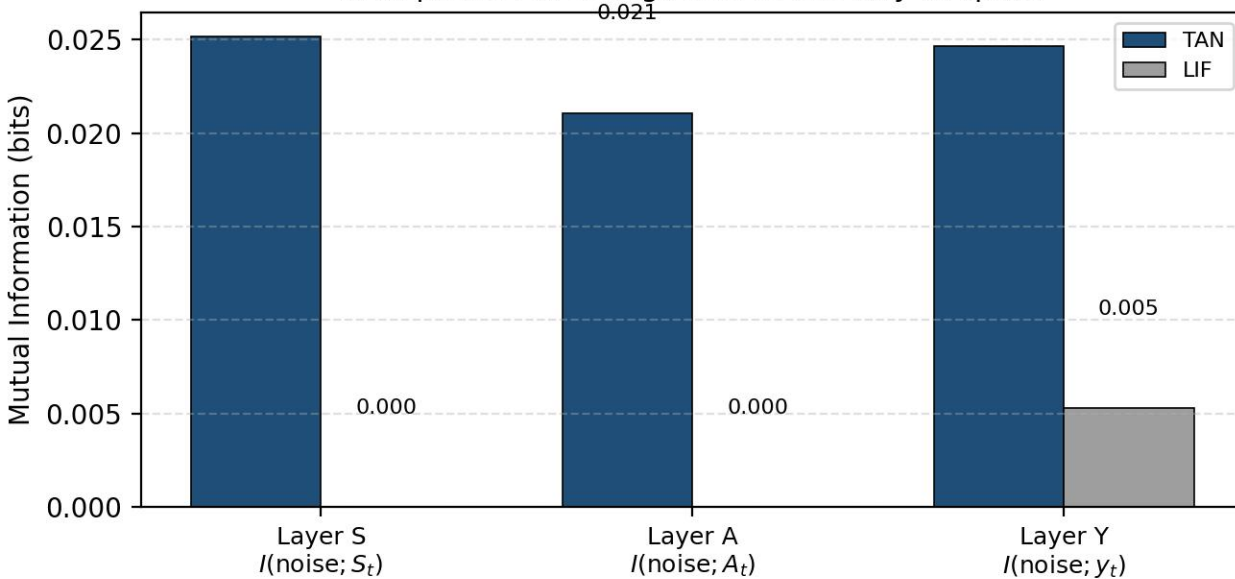
(a) Three-Layer Information Cascade: TAN vs LIF



(b) Noise Information Cascade
(TAN filters noise at the threshold; LIF does not)



(c) Clean Change-Rate Information Cascade
(TAN preserves change info all the way to spike)



(d) MI vs Noise Level
Internal MI stays high; spike-output noise MI stays low

